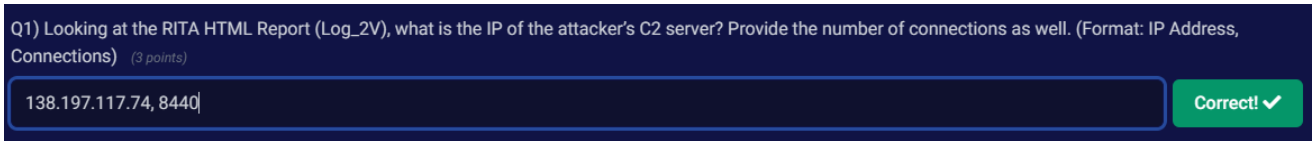
Nano

Analyst : Benjamin Liu

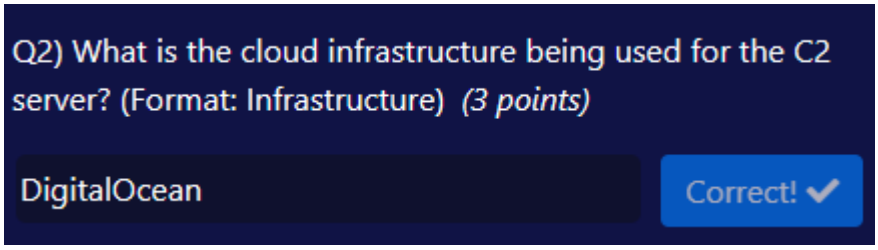
Date : 4/5/2026

- **Analyse Summary:**
  - An attack of malware **dnscat2** detected.
  - **high-frequency DNS queries, including 30189 subdomain of nanobotninjas.com and 82920 DNS requests.**
  - Encodes data inside **subdomain labels, like** 03b001050c0d05fbab, as payloads.
  - The malware uses both **A records + TXT records** for command exchange. By analyzing the usually high volumn of DNS connections carrying encoded payloads, we were abe to identify the attacker's C2 server IP address.
  - Produces **long, random-looking subdomains, such as** 03b001050c0d05fbab.cat.nanobotninjas.com, which is one of subdomain built for DNS tunnel.
- **Tools:**
  - **Zeek : A** powerful **framework for network traffic analysis and security monitoring. [** https://github.com/zeek/zeek **]**
  - **RITA** (Real Intelligence Threat Analytics): an open source framework for network traffic analysis. [FQDN (Fully Qualified Domain Name)]
- **Screenshots and Evidences**



In the RITA HTML_Report(LOg_2V), we notice that the connections are far more than others.

Score	Source	Destination	Connections	Avg. Bytes	Intvl. Range	Size Range	Intvl. Mode	Size Mode	Intvl. Mode Count	Size Mode Count	Intvl. Skew	Size Skew	Intvl. Dispersion	Size Dispersion	Total Bytes
0.996	10.234.234.100	138.197.117.74	8440	1317.000	8	935	10	544	7306	8312	0.000	0.000	0	0	11115975



We have known the C2 server IP. Then We search the ip at Ipinfo.io.

138.197.117.74

Star

Clifton, New Jersey, US | Cloud | Hosting | Webserver

- Summary
- Geolocation
- Privacy
- ASN
- Company
- Abuse

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Summary

ASN	AS14061 - DigitalOcean, LLC
Hostname	No Hostname
Range	138.197.112.0/20
Company	DigitalOcean, LLC
Hosted domains	0
Privacy	True
Anycast	False
ASN type	Hosting
Abuse contact	abuse@digitalocean.com

Q3) Looking at the RITA User Agent report, what is the system that corresponds to the connection count in Q1? (Format: User Agent String) *(3 points)*

Mozilla/5.0 (compatible; MSIE 9.0; Windows

Correct! ✔

Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; Trident/5.0)	8439

Q4) Let's look at Log_1D, what is the low-profile subdomain with the absurd amount of requests? Provide the number of requests as well. (Format: Subdomain, Request) *(3 points)*

cat.nanobotninjas.com, 82920

Correct! ✔

file:///home/ubuntu/Desktop/Investigation/RITA HTML Report/Log_1D/dns.html

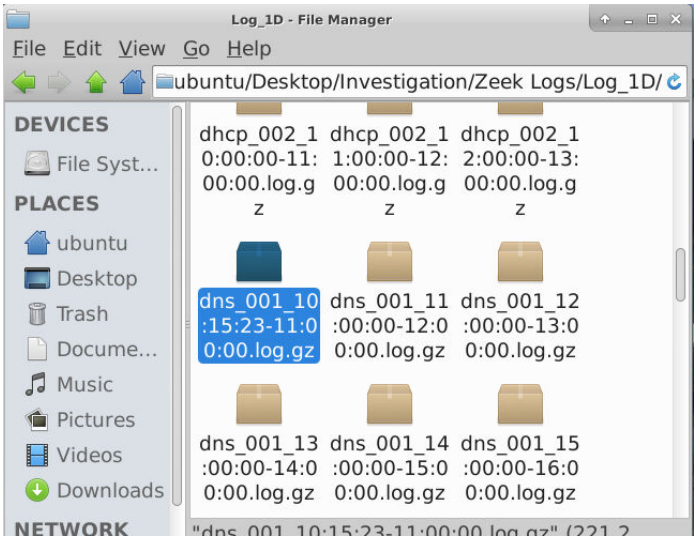
RITA | Viewing: Log_1D | Beacons | Beacons FQDN | Strobes | DNS | BL Source IPs | BL Dest. IPs | BL

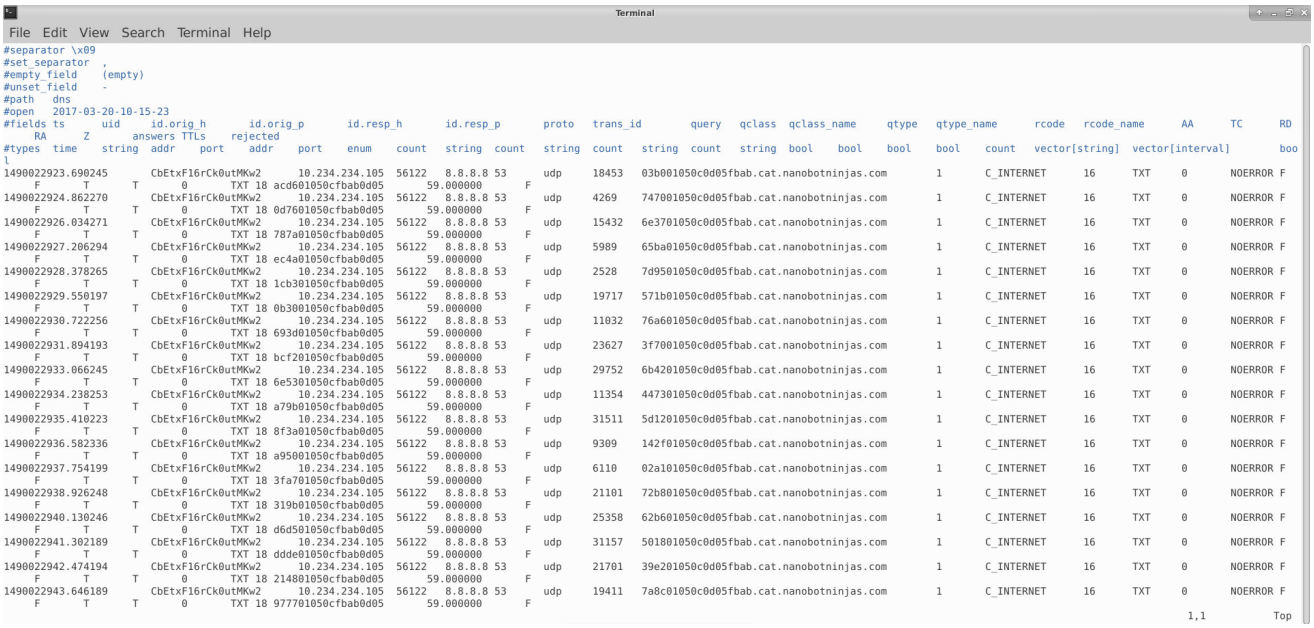
Subdomain Count	Visited	Domain
30189	82920	nanobotninjas.com
30189	82920	cat.nanobotninjas.com

Q5) In the Zeek logs (Log_1D), we can see a large number of DNS TXT record requests for a private IP for the subdomain found in Q4. List the IP. (Format: IP Address) *(3 points)*

10.234.234.105 Submit

Let us open the path of Zeek in the incestigation filefold. And locate the DNS logs. After





So we get the IP address.

Q6) This is an unusual activity. To avoid cached results on the intermediate DNS server(s), a certain value prepended to the subdomain is being changed. What numeral system is being used for the value that changes? It seems like a Base 16 numbering system. (Format: value) *(5 points)*

hex Submit

"Base16", "16" are not right answer.

```
03b001050c0d05fbab.cat.nanobotninjas.com
747001050c0d05fbab.cat.nanobotninjas.com
6e3701050c0d05fbab.cat.nanobotninjas.com
65ba01050c0d05fbab.cat.nanobotninjas.com
7d9501050c0d05fbab.cat.nanobotninjas.com
571b01050c0d05fbab.cat.nanobotninjas.com
76a601050c0d05fbab.cat.nanobotninjas.com
3f7001050c0d05fbab.cat.nanobotninjas.com
6b4201050c0d05fbab.cat.nanobotninjas.com
447301050c0d05fbab.cat.nanobotninjas.com
5d1201050c0d05fbab.cat.nanobotninjas.com
142f01050c0d05fbab.cat.nanobotninjas.com
02a101050c0d05fbab.cat.nanobotninjas.com
```

Q7) Judging from the logs (Log_1D), what tool was the attacker using? A tool to help C2s channel over the DNS protocol. (Format: tool) *(5 points)*

Answer: dnscat2

- **dnscat2** is a well-known tool specifically designed to create **C2 channels over DNS**, and its traffic patterns match exactly what you described
  - Uses **high-frequency DNS queries**
  - Encodes data inside **subdomain labels**
  - Uses **A records + TXT records** for command exchange
  - Produces **long, random-looking subdomains**
  - Is widely used in training labs and red-team exercises